Open problems:
Inconsistent convergence in Hilbert space

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked *research* are open research questions, not needed to check the brief.

From the brief

1. **Unbounded observables in general.** The introduction states that the domain of an unbounded self-adjoint operator cannot be the whole space, and that every unbounded operator is "not continuous in its own domain", e.g. for the average position. Only the single example $\langle N \rangle$ on ℓ^2 is formalized; the general statements (Hellinger–Toeplitz, discontinuity of $\psi \mapsto \langle \psi | A | \psi \rangle$ for every unbounded A) are not. (Brief, Sec. 1.)

2. **Schwartz topology prevents it.** "A topology that keeps track of all moments, like the Schwartz topology, would not allow this kind of convergence." That $\langle N \rangle$, and expectations of polynomial observables in general, are continuous on $S(\mathbb{N})$ is not proved. (Brief, Sec. 3, last paragraph.)

3. **A criterion of physical convergence.** *Research.* The brief concludes that the ℓ^2 topology "is too weak to take into account all conditions for a meaningful criterion of physical convergence". Which topology is the right one, and whether it is the Schwartz topology, is left open. (Brief, Sec. 4.)